

Nonnegative and positive factorizations

The Rowland–Wu polynomial identity for Sinkhorn limits

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Topic: Exact matrix scaling and balancing.

Rating rationale: Challenging because the full coefficient formula requires a general algebraic-combinatorial identity; community importance concerns exact formulas for a widely used matrix-scaling limit.

Resolution — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** **PASS for the exact target.**

The complete Rowland–Wu coefficient identity holds for every positive real rectangular matrix and all $m, n \geq 1$. The proof identifies the coefficient sum with one determinant and constructs a null vector after scaling. Vanishing minors and the cases $m = 1$ or $n = 1$ are included. This proves the displayed coefficients, beyond the previously known algebraic-degree bound.

The complete target is resolved. Its former difficulty rating is historical; the original statement, references and dated audits remain below.

Primary reference: [complete authored PDF](#), [standalone TeX](#), **Theorem 1**. [Independent proof review](#) · [Authorship and submission record](#). The 11 September verification was independent agent review. The later Lean verification is recorded below; no external human peer review is claimed.

Lean proof and verification evidence

The complete original target is Lean verified, 17 September 2026. The [immutable proof](#) at revision `21ed3545a8b4784303e9cc0473879eb33ef4d813` passed [non-root Linux run 35276203784](#). The [retained execution evidence](#) authenticates all 380 submitted project inputs and 35 exported statements. Comparator statement matching, default-kernel replay, standard transitive axioms, sandbox checks and rejection controls passed.

The main declaration `NLA.NM04.rowland_wu_identity` proves the literal subset-sum coefficient formula for every strictly positive real rectangular matrix with both dimensions at least one. The formalization also proves positive scaling existence and uniqueness, the four transition signs, sorted indices, scaling covariance, and empty or singular minor cases. The [35 contracts](#), [implementation map](#) and [source correspondence](#) identify the full target. They use the original equivalent scaling definition; convergence of a particular normalization algorithm is outside this claim.

Formalization contributor: **George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology**. Matthew J. Colbrook retains mathematical solution credit; Eric Rowland and Jason Wu retain the question. Two independent statement reviews preceded implementation, and two independent nonauthor final source reviews approved the proof. [Review records](#) disclose substantial OpenAI Codex assistance and exact reviewer scopes.

The project pins Lean 4.33.1, Mathlib and LeanCert. Two exact half inequalities execute through LeanCert in kernel mode and are consumed by coercivity; dimensions, minors and sums remain symbolic. Every export uses only `propext`, `Classical.choice` and `Quot.sound`. Run `lake build` in the [Lean project](#); its reproduction instructions distinguish local compilation from the Linux checker.

Problem statement

Let $m, n \geq 1$ and $A \in \mathbb{R}_{>0}^{m \times n}$. Let $\text{Sink}(A)$ be its Sinkhorn limit with row sums 1 and column sums m/n : equivalently, the unique matrix $D_1 A D_2$ with these sums, where D_1, D_2 are positive diagonal matrices. Set $x = \text{Sink}(A)_{11}$.

Define

$$\mathcal{D} = \{(R, C) : R \subseteq \{2, \dots, m\}, C \subseteq \{2, \dots, n\}, |R| = |C|\}.$$

Submatrix indices are in increasing order; empty determinants and products equal 1. Put

$$\Delta(R, C) = \det A_{\{1\} \cup R, \{1\} \cup C}, \quad \Gamma(R, C) = a_{11} \det A_{R, C},$$

$$M(\mathcal{S}) = \prod_{(R, C) \in \mathcal{S}} \Delta(R, C) \prod_{(R, C) \in \mathcal{D} \setminus \mathcal{S}} \Gamma(R, C).$$

For a finite set U of integers and $s \in U$, let $p_s(U)$ be the position of s in the increasing list of U , starting at 1. For each $\mathcal{S} = \{(R_i, C_i) : 1 \leq i \leq k\} \subseteq \mathcal{D}$, define $H_{\mathcal{S}} \in \mathbb{Z}^{k \times k}$ by

$$(H_{\mathcal{S}})_{ii} = |R_i|(m+n) - mn, \quad \tau_{ij} = (R_i \setminus R_j, R_j \setminus R_i, C_i \setminus C_j, C_j \setminus C_i),$$

and, for $i \neq j$,

$$(H_{\mathcal{S}})_{ij} = \begin{cases} (-1)^{p_s(R_j) + p_t(C_j)} m, & \tau_{ij} = (\emptyset, \{s\}, \emptyset, \{t\}), \\ (-1)^{p_s(R_i) + p_t(C_i) + 1} n, & \tau_{ij} = (\{s\}, \emptyset, \{t\}, \emptyset), \\ (-1)^{p_s(C_i) + p_t(C_j)} m, & \tau_{ij} = (\emptyset, \emptyset, \{s\}, \{t\}), \\ (-1)^{p_s(R_i) + p_t(R_j)} n, & \tau_{ij} = (\{s\}, \{t\}, \emptyset, \emptyset), \\ 0, & \text{otherwise.} \end{cases}$$

The determinant is independent of the ordering chosen for $\mathcal{S}$. Is the following identity valid for every such A, m, n ?

$$\sum_{\mathcal{S} \subseteq \mathcal{D}} \det(m^{-1} H_{\mathcal{S}}) M(\mathcal{S}) x^{|\mathcal{S}|} = 0.$$

Why it matters

This would give an explicit algebraic relation for entries of a ubiquitous iterative matrix scaling limit, including a structured formula for every coefficient.

References

- E. Rowland and J. Wu, [The entries of the Sinkhorn limit of an \$m \times n\$ matrix](#), arXiv v2 (2025-05-25), notation on pp. 3–4 and Conjecture 2. The matrix denoted H_Δ here is their $\text{adj}_\Delta(m, n)$.
- E. Rowland, [Combinatorial structure behind Sinkhorn limits](#), SIAM talk, 2025-07-07, slide 13: the coefficient formula remains conjectural after the degree bound was proved.
- M. C. Fang, [Closed Form of a Generalized Sinkhorn Limit](#), 2025, abstract and the general algebraic degree bound. This settles the degree-bound consequence, rather than the displayed coefficient formula.

Status check — 2026-09-10

Rechecked [Rowland–Wu v2, §2 and Conjecture 2](#), including the rectangular scaling and four coefficient signs, and [Fang’s degree-bound result](#). The 3×3 coefficient formula is proved, while the determinant formula in arbitrary dimensions remains conjectural. Searches for subsequent Rowland–Wu identity proofs found no full resolution. Fang’s general degree bound alone does not establish the displayed coefficients.